

Md Zesun Ahmed Mia

R&D Device Engineer | Next-Generation Memory & ML Hardware

A dedicated device engineer with a track record of innovation in spintronics and neuromorphic hardware. Experienced in the full R&D cycle, from concept and fabrication to testing and analysis.

Education

- 08/2022–05/2027 **Ph.D. in Electrical Engineering**, *Pennsylvania State University*, University Park, PA, USA, CGPA: 4.00/4.00
(Expected) Advisor: Dr. Abhronil Sengupta
- 08/2022–08/2024 **M.S. in Electrical Engineering**, *Pennsylvania State University*, University Park, PA, USA, CGPA: 4.00/4.00 | Thesis: Neuromorphic Computing for Lifelong Learning
- 02/2015–04/2019 **B.Sc. in Electrical and Electronic Engineering**, *Bangladesh University of Engineering & Technology (BUET)*, Dhaka, Bangladesh, CGPA: 3.81/4.00

Industrial Experience

- 05/2025–07/2025 **Graduate Technical Intern**, *Intel Corporation*, Hillsboro, OR
- Led thin film deposition process development for advanced memory and logic device fabrication at cutting-edge technology nodes, with specific focus on memory applications including DRAM, SRAM, and emerging memory technologies.
 - Applied AI-driven analysis to correlate material properties with device performance, enabling data-driven device design optimization.
 - Investigated novel process integration approaches for next-generation device architectures, supporting enhanced device reliability and performance.
- 09/2020–07/2021 **R&D Engineer**, *SEMWAVES Ltd.*, London, UK, part-time
- Delivered a 50 kW hybrid renewable system (solar + smallscale hydro) for an offgrid Bangladeshi site, supporting reliable community power.
 - Owned technical leadership and coordination (vendors, field teams, stakeholders); directed device selection, integration, QA/safety procedures, and optimization against load profiles and uptime targets.

Technical Skills

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| Device R&D, Modeling | Spintronic devices, NVM, advanced memory, SRAM/DRAM cell design and characterization, PIM, near-memory computing, memory-centric computing, process integration, TCAD, compact modeling, parameter extraction, Verilog-A |
| EDA/Sim. | TCAD, Cadence Virtuoso, Spectre, HSPICE, COMSOL, ModelSim, Synopsys (Design Compiler, PrimeTime, VCS), CIM (Compute-In-Memory) Systems |
| Programming | Python, CUDA, MATLAB, Verilog, Shell, C/C++, Bash, Linux/Unix |
| ML/AI Tools | PyTorch, TensorFlow, TensorRT, Data Visualization (Matplotlib, Seaborn, Plotly), Pandas, NumPy, JMP, Jupyter, LaTeX, AI accelerator development, Systolic Arrays, HPC applications, parallel computing architectures |
| AI Dev. | Generative and agentic AI tools (Cursor, Copilot, etc.) for research, design, and code development |
| Hardware | Device-circuit co-design, PCB design, Oscilloscope, Signal Generator, LabVIEW |
| Collaboration | Git, Microsoft Office, Google Workspace |
| Fabrication | Lithography, Etching, Deposition (CVD, HDP-CVD, PVD), Cleanroom, E-beam |
| Characterization & Reliability | AFM, SEM, Probe Station, Electrical Testing (DC/AC characterization, IV/CV measurements), TEM, endurance, retention, failure analysis, Metrology (XRR, XRD, ellipsometry, profilometry) |

Academic Research and Teaching Experience

- 08/2022– **Graduate Research Assistant**, *Penn State*, University Park, PA
- Present
- Developed and optimized fabrication recipes for spintronic devices, successfully fabricating and characterizing Hall bar structures in a university cleanroom environment.
 - Designed and simulated advanced FET architectures using TCAD (Silvaco/Sentaurus) and developed physics-based models for FeFETs in MATLAB from experimental characterization data.
 - Utilized experimental characterization data of Ferroic devices (e.g. FeFETs) to inform device models and algorithm-hardware co-design for next-generation memory and logic applications.
- 08/2024– **Graduate Teaching Assistant**, *Penn State*, State College, PA
- 05/2025
- Taught Cadence Virtuoso (schematic/layout), PDK usage, DRC/LVS, and analog/digital design flows; created hands-on lab modules and guided tool/debug workflows.
 - Supervised Capstone projects for 90+ undergraduate students across communications, electronics, and firmware: supported Raspberry Pi/Arduino development, software-defined radio experiments, PCB design, and system integration end-to-end.
- 02/2021– **Lecturer**, *University of Liberal Arts Bangladesh (ULAB)*, Dhaka, Bangladesh
- 08/2022
- Taught undergraduate courses: Solid State Devices, Digital Circuit Design, Semiconductor Device Physics, Power Electronics.
 - Developed lab modules and supervised projects on semiconductor devices and circuits.

Projects

- Astromorphic Transformer** Lead Student Researcher, 2022–2025. Developed a bioplausible transformer architecture leveraging neuron-astrocyte interactions to emulate self-attention mechanisms. Incorporated Hebbian and presynaptic plasticities with non-linearities and feedback, achieving superior accuracy and faster convergence on sentiment classification (IMDB), image classification (CIFAR-10), and language modeling (WikiText-2) tasks. See publication: [IEEE TCDS 2025].
- Spintronic Device-Based Memory** Lead Student Researcher, 2023–Present. Led end-to-end fabrication of spintronic memory arrays using e-beam lithography, sputtering deposition, and lift-off patterning. Performed comprehensive electrical characterization (I-V, R-H loops, retention, endurance) and extracted device parameters for compact model development. Advanced non-volatile memory technologies for neuromorphic computing and in-memory processing applications.
- Ultra Low Cost Electronic Braille Device** First Author, 2022–2024. Designed and fabricated a portable, solenoid-based electronic Braille device using 3D printing and Arduino microcontroller. Achieved ultra-low cost (\$20) and lightweight design (338g) with 100 frames/s refresh rate, enabling real-time text-to-braille translation. Implemented PWM control for low power consumption and USB/Bluetooth connectivity. See publication: [iCACCESS 2024].

Select Publications

- [1] **Md Zesun Ahmed Mia**, Malyaban Bal, and Abhronil Sengupta. “Delving deeper into astromorphic transformers”. In: *IEEE Transactions on Cognitive and Developmental Systems* (2025).
- [2] Arnob Saha, **Md Zesun Ahmed Mia**, Jiahui Duan, Kai Ni, and Abhronil Sengupta. “Toward Variation-Tolerant Ferroelectric Neural Computing: Special Session Paper”. In: *2025 IEEE 68th International Midwest Symposium on Circuits and Systems (MWSCAS)*. IEEE. 2025, pp. 583–587.
- [3] Tao Zhang, Mingjie Hu, **Md Zesun Ahmed Mia**, Hao Zhang, Wei Mao, Katsuyuki Fukutani, Hiroyuki Matsuzaki, Lingzhi Wen, Cong Wang, Hongbo Zhao, et al. “Self-sensitizable neuromorphic device based on adaptive hydrogen gradient”. In: *Matter* 7.5 (2024), pp. 1799–1816.
- [4] KM Ashraful Hoque Fahim, Md Jubair Hasan Khalid, **Md Zesun Ahmed Mia**, and Mirza Rasheduzzaman. “Study of 3-nm Cylindrical GAAFETs with Variations in High-k Dielectric Gate-oxide Materials”. In: *2022 IEEE Symposium on Industrial Electronics & Applications (ISIEA)*. IEEE. 2022, pp. 1–5.

Recognitions

- The Wormley Family Graduate Fellowship, Harry G. Miller Fellowships in Engineering (2025)
- Arthur Waynick Graduate Scholarship (2024), Milton and Albertha Langdon Memorial Fellowship (2023), Melvin P. Bloom Memorial Fellowship (2022)

Professional Affiliations

- Reviewer, IEEE TNNLS (2025), Design Automation Conference (DAC) (2025), IEEE MWSCAS (2025)
- Student Member, IEEE (2015–Present), Executive Member, EDS, IEEE BD (2021–2022)